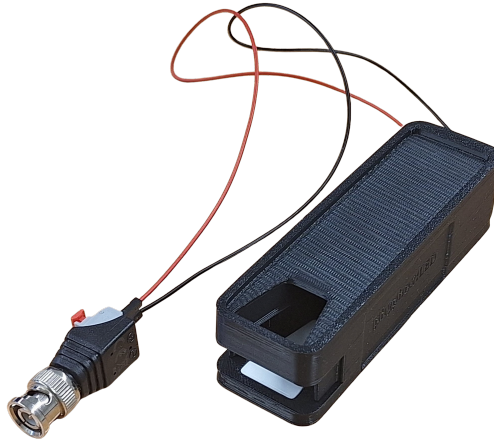


Planck's Constant

Material



phyphox:h-box (voltage control)



Blackout box



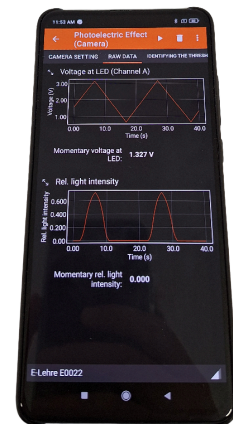
Stabilisation aid



Various LED inserts of different wavelengths



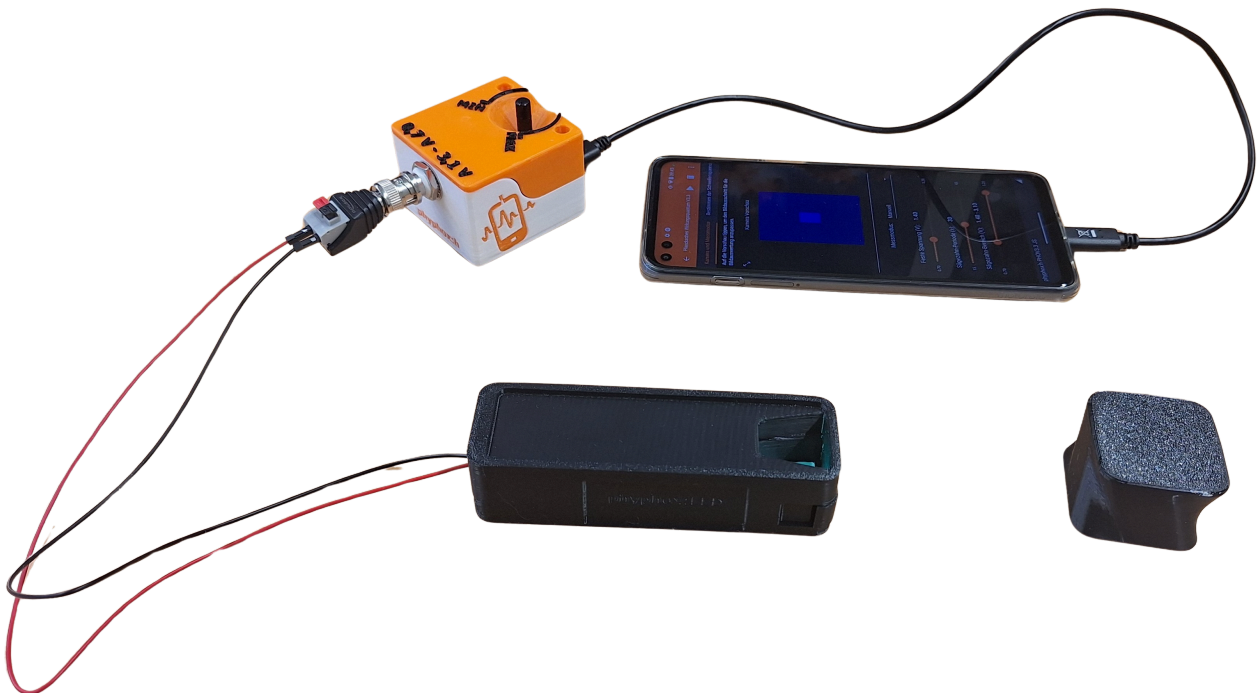
Connection cable



Smartphone (light intensity measurement)

Setup/Preparation

1. Set up the experimental apparatus according to the figure below.
2. Download the corresponding *phyphox* experiment bundle:
 - The *phyphox*: h-Box must be connected to the smartphone (or another power source).
 - Open the *phyphox* app.
 - Click on the plus symbol.
 - Select „Add experiment for Bluetooth device“.
 - Select your *phyphox*:h-Box from the list (numbering on the side).
 - Wait until the download is complete.
 - Save the desired experiment (or both) in the collection.



Completed setup

Procedure with the smartphone camera

- If the appropriate *phyphox* experiment does not open automatically or needs to be reopened during the experiment:
 1. Select the *phyphox* experiment **Planck's Constant** (camera in the icon) under the *Quantum Physics* section.
 2. Select your *phyphox*: h-Box from the appearing list of Bluetooth devices (the number is on the side of the box). $\Rightarrow$ The box will now connect to the app and send the value of the applied voltage.
- Determine the threshold voltages of several LEDs (13 steps):
 1. Insert an LED insert into the blackout box.
 2. Set the voltage regulator on the *phyphox*: h-Box to maximum.
 3. Place your smartphone on the blackout box so that the camera is positioned above the white reflective material. Use the stabilization aid if necessary.
 4. Reduce the voltage to the highest value at which the camera's selection area *has no overexposure anywhere* (i.e. the selection area must *not contain any red-marked areas*).
 - The selection area is the slightly brighter rectangle within the camera preview.
 - Overexposure may occur outside the selection area.
 - The selection area can be moved. To do this, tap on the camera preview.
 5. Switch to the „Identify the threshold voltage“ tab.
 6. Start the measurement (play button in the top right).
 7. Record an LED characteristic curve by *slowly* reducing the voltage.
 8. Pause the measurement.
 9. Place the green regression line in the characteristic curve *by adjusting the minimum and maximum considered intensity*. The regression line should run through the approximately linear region of the characteristic curve. Any saturation curve should therefore *not* be taken into account.
 10. Note down the threshold voltage $U_{\text{threshold}}$ calculated by the experiment in the table on page 7. Also note the corresponding uncertainty.

11. Enter the wavelength of the LED in the experiment and save the *wavelength-threshold voltage* pair in the experiment. The pair is now available in the „Measuring h“ tab.
12. Do *not* use the button with the trash can symbol during the procedure. This deletes *all* data.
Now delete the measurement curve using the button provided specifically for this purpose.
13. Repeat the measurement process with the next LED.

Procedure without the smartphone camera

- If the appropriate *phyphox* experiment does not open automatically or needs to be reopened during the experiment:
 1. Select the *phyphox* experiment **Planck's Constant** (no camera in the icon) under the *Quantum Physics* section.
 2. Select your *phyphox*: h-Box from the appearing list of Bluetooth devices (the number is on the side of the box). $\Rightarrow$ The box will now connect to the app and send the value of the applied voltage.
- Determine the threshold voltages of several LEDs (11 steps):
 1. Set the voltage regulator on the *phyphox*: h-Box to minimum.
 2. Insert an LED insert into the blackout box.
 3. Switch to the „Determine Threshold Voltage“ tab.
 4. Start the measurement (play button in the top right).
 5. Record an LED characteristic curve by varying the voltage.
 6. Pause the measurement.
 7. Place the green regression line in the characteristic curve *by adjusting the minimum and maximum considered current*. The regression line should run through the approximately linear region of the characteristic curve. Any saturation curve should therefore *not* be taken into account.
 8. Note down the threshold voltage $U_{\text{threshold}}$ calculated by the experiment in the table on page 7. Also note the corresponding uncertainty.
 9. Enter the wavelength of the LED in the experiment and save the *wavelength-threshold voltage* pair in the experiment. The pair is now available in the „Measuring h“ tab.
 10. Do *not* use the button with the trash can symbol during the procedure. This deletes *all* data.
Now delete the measurement curve using the button provided specifically for this purpose.
 11. Repeat the measurement process with the next LED.

Analysis

1. Calculate the corresponding light frequencies with corresponding uncertainty from the wavelengths entered in the table on page 7. The uncertainty is determined using Gaussian error propagation.
2. Calculate the corresponding photon energies with their respective uncertainties from the threshold voltages.
3. Plot the pairs using Python or Excel (including error bars) and create a regression line.
4. Draw the lines of minimum and maximum slope through the uncertainty regions.
5. Calculate a value for Planck's constant including uncertainty from the slopes of the three drawn lines.
6. Perform a comparison of the obtained value with the literature value.

Table 1: Measurement data of threshold voltage as a function of wavelength.

λ [nm]	u_λ [nm]	f [10^{14} Hz]	u_f [10^{14} Hz]	$U_{\text{threshold}}$ [V]	u_U [V]	E_{photon} [J]	u_E [J]
405	7						
463	8						
469	10						
504	18						
575	8						
586	8						
591	7						
595	8						
630	5						
631	8						
655	9						